

COR-STAT1202 Introductory Statistics

Tutorial 4

Discrete Probability Distributions

Q1. A car salesperson estimated the following probabilities for the number of cars that she will sell in the next week:

Number of Cars	0	1	2	3	4	5
Probability	0.1	0.2	0.35	0.16	0.12	0.07

- Find the expected number of cars that will be sold in the week.
- Find the standard deviation of the number of cars that will be sold in the week.
- The salesperson receives a salary of \$250 for the week, plus an additional \$300 for each car sold. Find the mean and standard deviation of her total salary for the week.
- What is the probability that the salesperson's salary for the week will be more than \$1,000?

Answer:

Cars	P(x)	xP(x)	(x-μ) ² P(x)
0	0.1	0	0.48841
1	0.2	0.2	0.29282
2	0.35	0.7	0.015435
3	0.16	0.48	0.099856
4	0.12	0.48	0.384492
5	0.07	0.35	0.544887
	Sum =	2.21	1.8259
		Std Dev, σ =	1.35125867

a) $E(X) = \sum [x P(x)] = 2.21$

b) Variance, $\sigma^2 = \sum [(x - E(X))^2 P(x)] = 1.8259$

Standard deviation, $\sigma = 1.3513$

c) $E(a + bX) = a + b\mu_X$

Mean Salary = \$250 + \$300 (2.21) = \$913.

$Var(a + bX) = b^2 \sigma_X^2 \Rightarrow \sigma = b\sigma_X$

Standard deviation of salary = \$300(1.3513) = \$405.39

- d) \$250+3(\$300) = \$1,150 > \$1,000 => to earn a salary of \$1,000 or more, the salesperson must sell at least 3 cars.

$P(X \geq 3) = 0.16 + 0.12 + 0.07 = 0.35$

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Q2. A small commuter airline flies planes that can seat up to 8 passengers. The airline has determined that the probability that a passenger with ticket will not show up for a flight is 0.2. For each flight the airline sells tickets to at most the first 10 people placing orders.

The probability distribution for the number of tickets sold per flight is shown in the accompanying table.

Number of tickets sold/flight	6	7	8	9	10
Probability	0.25	0.35	0.25	0.10	0.05

- Determine the Probability of 6, 7, 8 and 9 passengers with tickets showing up when 9 tickets were sold.
- Determine the Probability of 6, 7, 8, 9 and 10 passengers with tickets showing up when 10 tickets were sold.
- For what proportion of the airline's flights does the number of passengers with ticket showing up exceed the number of available seats? (Assume independence between the number of tickets sold and the probability that a passenger with ticket will show up (Hint: $P(A \cap B) = P(A) P(B)$)

Answer:

Finding the probability of overbooking a flight. Let x be the no. of passengers showing up.

The probability of a passenger with ticket showing up for a flight is, $p = 1 - 0.2 = 0.8$. The probabilities of the binomial distribution are shown below.

x	a) Probability of no. of passengers showing up (when 9 tickets were sold)	b) Probability of no. of passengers showing up (when 10 tickets were sold)
6	$C_6^9 (0.8)^6 (0.2)^3 = 84(0.8)^6 (0.2)^3 = 0.17617$	$C_6^{10} (0.8)^6 (0.2)^4 = 210(0.8)^6 (0.2)^4 = 0.08808$
7	$C_7^9 (0.8)^7 (0.2)^2 = 36(0.8)^7 (0.2)^2 = 0.30199$	$C_7^{10} (0.8)^7 (0.2)^3 = 120(0.8)^7 (0.2)^3 = 0.20133$
8	$C_8^9 (0.8)^8 (0.2)^1 = 9(0.8)^8 (0.2)^1 = 0.30199$	$C_8^{10} (0.8)^8 (0.2)^2 = 45(0.8)^8 (0.2)^2 = 0.30199$
9	$C_9^9 (0.8)^9 (0.2)^0 = 1(0.8)^9 (0.2)^0 = 0.13422$	$C_9^{10} (0.8)^9 (0.2)^1 = 10(0.8)^9 (0.2)^1 = 0.26844$
10		$C_{10}^{10} (0.8)^{10} (0.2)^0 = 1(0.8)^{10} (0.2)^0 = 0.10737$

- Since 10% of the time 9 tickets are sold and 5% of the time 10 tickets are sold, the proportion of flights where the number of passengers with ticket showing up exceeds the number of available seats (i.e. 8 seats) is:

$$\begin{aligned}
 & P(\text{flights with 9 tickets sold and 9 passengers show up}) + P(\text{flights with 10 tickets sold and 9 passengers show up}) + P(\text{flights with 10 tickets sold and 10 passengers show up}) \\
 &= (0.10) P(x=9) + (0.05) P(x=9) + (0.05) P(x=10) \\
 &= (0.10) (0.13422) + (0.05) (0.26844) + (0.05) (0.10737) \\
 &= 0.0322
 \end{aligned}$$

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Q3. Records indicate that, on average, 3.2 breakdowns per day occur on an urban highway during the morning rush hour. Assume that breakdown rate follows a Poisson distribution.

- a) Find the probability that on any given day there will be fewer than 2 breakdowns on this highway during the morning rush hour.
- b) Find the probability that on any given day there will be more than 4 breakdowns on this highway during the morning rush hour

Answer:

$$\begin{aligned}\text{a) } P(x < 2) &= P(x \leq 1) \\ &= P(x = 0) + P(x = 1) \\ &= (e^{-3.2}) \frac{3.2^0}{0!} + (e^{-3.2}) \frac{3.2^1}{1!} \\ &= 0.04076 + 0.13044 \\ &= 0.1712\end{aligned}$$

$$\begin{aligned}\text{b) } P(x > 4) &= 1 - P(x \leq 4) \\ &= 1 - P(x = 4) - P(x = 3) - P(x = 2) - P(x = 1) - P(x = 0) \\ &= 1 - 0.17809 - 0.22262 - 0.20870 - 0.04076 - 0.13044 \\ &= 1 - 0.7806 \\ &= 0.2194\end{aligned}$$

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Q4. For credit card holders with one to three cards, the joint probabilities for number of cards owned (X) and the number of credit purchases made in a week (Y) is as shown below. For a random chosen person from this group,

- what is the probability distribution for number of purchases made in a week?
- what is the probability distribution for number of purchases made in a week if this person has three cards?
- are number of cards owned and number of purchases made statistically independent?

Number of Cards (X)	Number of Purchases (Y)				
	0	1	2	3	4
1	0.08	0.13	0.09	0.06	0.03
2	0.03	0.08	0.08	0.09	0.07
3	0.01	0.03	0.06	0.08	0.08

Answer:

a) $P(y) = \sum_x P(x, y)$

$$P(y = 0) = 0.08 + 0.03 + 0.01 = 0.12$$

$$P(y = 1) = 0.13 + 0.08 + 0.03 = 0.24$$

$$P(y = 2) = 0.09 + 0.08 + 0.06 = 0.23$$

$$P(y = 3) = 0.06 + 0.09 + 0.08 = 0.23$$

$$P(y = 4) = 0.03 + 0.07 + 0.08 = 0.18$$

b) $P(y=0|x=3) = \frac{P(y=0 \cap x=3)}{P(x=3)} = \frac{0.01}{0.26} = \frac{1}{26} = 0.0385$

$$P(y=1|x=3) = \frac{P(y=1 \cap x=3)}{P(x=3)} = \frac{0.03}{0.26} = \frac{3}{26} = 0.1154$$

$$P(y=2|x=3) = \frac{P(y=2 \cap x=3)}{P(x=3)} = \frac{0.06}{0.26} = \frac{6}{26} = 0.2308$$

$$P(y=3|x=3) = \frac{P(y=3 \cap x=3)}{P(x=3)} = \frac{0.08}{0.26} = \frac{8}{26} = 0.3077$$

$$P(y=4|x=3) = \frac{P(y=4 \cap x=3)}{P(x=3)} = \frac{0.08}{0.26} = \frac{8}{26} = 0.3077$$

c) $P(x = 3 \cap y = 4) = 0.08$

$$P(x = 3) P(y = 4) = 0.26(0.18) = 0.0468$$

$$P(x = 3 \cap y = 4) \neq P(x = 3) P(y = 4)$$

$\Rightarrow$ number of cards owned and number of purchases made are not statistically independent.

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Q5. A restaurant manager receives occasional complaints about the quality of both the food and the service. The marginal probability distributions for the number of weekly complaints are shown in the table below. Assume that the complaints about the food and service are independent of each other.

No. of food complaints	Probability	No. of service complaints	Probability
0	0.12	0	0.18
1	0.29	1	0.38
2	0.42	2	0.34
3	0.17	3	0.10

- Find the joint probability distribution.
- Find the mean and variance of the total number of complaints received in a week.

Answer:

a)

Because of independence, the joint probabilities are the products of the marginal probabilities, so $P(0,0) = 0.12(0.18) = 0.0216$, and so on.

		Y Service				
	X Food	0	1	2	3	P(x)
	0	0.0216	0.0456	0.0408	0.012	0.12
	1	0.0522	0.1102	0.0986	0.029	0.29
	2	0.0756	0.1596	0.1428	0.042	0.42
	3	0.0306	0.0646	0.0578	0.017	0.17
	P(y)	0.18	0.38	0.34	0.1	1

b)

$$E(X) = \sum [x P(x)] = 0(0.12) + 1(0.29) + 2(0.42) + 3(0.17) = 1.64$$

$$E(Y) = \sum [y P(y)] = 0(0.18) + 1(0.38) + 2(0.34) + 3(0.10) = 1.36$$

$$\begin{aligned} Var(X) &= \sum [(x - E(X))^2 P(x)] \\ &= (0 - 1.64)^2 0.12 + (1 - 1.64)^2 0.29 + (2 - 1.64)^2 0.42 + (3 - 1.64)^2 0.17 \\ &= 0.8104 \end{aligned}$$

$$\begin{aligned} Var(Y) &= \sum [(y - E(Y))^2 P(y)] \\ &= (0 - 1.36)^2 0.18 + (1 - 1.36)^2 0.38 + (2 - 1.36)^2 0.34 + (3 - 1.36)^2 0.10 \\ &= 0.7904 \end{aligned}$$

For independent variables X and Y:

$$E(X + Y) = E(X) + E(Y) = 1.64 + 1.36 = 3$$

$$Var(X + Y) = Var(X) + Var(Y) = 0.8104 + 0.7904 = 1.6008$$

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Extra Questions

1. The following table shows the frequency distribution of X, a random variable representing the number of fish caught in a 3-hour period, determine the probability distribution of X. Calculate the expected value, variance, and standard deviation of X.

x	1	2	3
Frequency	9	7	4

Answer:

x	1	2	3
Probability	$\frac{9}{20} = 0.45$	$\frac{7}{20} = 0.35$	$\frac{4}{20} = 0.2$

$$E(X) = \sum [x P(x)] = 1(0.45) + 2(0.35) + 3(0.2) = 1.75$$

$$\begin{aligned} Var(X) &= \sum [(x - E(X))^2 P(x)] \\ &= (1 - 1.75)^2 0.45 + (2 - 1.75)^2 0.35 + (3 - 1.75)^2 0.2 \\ &= 0.2531 + 0.0219 + 0.3125 \\ &= 0.5875 \end{aligned}$$

$$standard\ deviation = \sqrt{0.5875} = 0.7665$$

2. The number of customers arriving at a 7-eleven outlet follows a Poisson distribution with an average of 5 arrivals per hour.
- What is the probability that 7 customers arrive at the 7-eleven outlet in a one-hour period?
 - What is the probability that at most 3 customers arrive at the 7-eleven outlet in a one-hour period?
 - What is the probability that between 1 and 4 customers, inclusively, arrive at the 7-eleven outlet in a one-hour period?

Answer:

Let X represent the number of people arriving per hour.

$$\begin{aligned} \text{a) } P(X = 7) &= (e^{-5}) \frac{5^7}{7!} \\ &= 0.1044 \end{aligned}$$

$$\begin{aligned} \text{b) } P(X \leq 3) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= 0.00674 + 0.03369 + 0.08422 + 0.14037 \\ &= 0.2650 \end{aligned}$$

$$\begin{aligned} \text{c) } P(1 \leq X \leq 4) &= P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) \\ &= 0.03369 + 0.08422 + 0.14037 + 0.17547 \\ &= 0.4338 \end{aligned}$$

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3. The following table lists the relative frequency distribution of the number of customers at the customer service counter of Ikea in an hour. If the customer service staff spends 5 minutes on each customer, what is the mean and standard deviation of the time the customer service staff has to spend serving the customers at the counter each hour?

x	2	3	4	5	6	7
$P(x)$	0.07	0.11	0.23	0.31	0.18	0.1

Answer:

Let $Y = 5X$.

Mean of X , $E(X) = \sum [x P(x)] = 4.72$

Mean of Y , $E(Y) = 5E(X) = 23.6$

$$\begin{aligned} \text{Var}(X) &= \sum [(x - E(X))^2 P(x)] = \sum [x^2 P(x)] - [E(X)]^2 \\ &= 24.08 - (4.72)^2 \\ &= 1.8016 \end{aligned}$$

$$\text{Var}(Y) = 5^2(\text{Var}(X))$$

$$\text{Standard deviation of } Y, \sigma_Y = \sqrt{5^2(1.8016)} = 6.7112$$

4. An airplane has four engines, each fails 11% of the time. Let X be the number of engines fail.
- Determine the probability distribution function of X .
 - Calculate the expected number and standard deviation of engine failures.

Answer:

a)

x	$P(x)$
0	$C_0^4(0.11)^0(0.89)^4 = 0.6274$
1	$C_1^4(0.11)^1(0.89)^3 = 0.3102$
2	$C_2^4(0.11)^2(0.89)^2 = 0.0575$
3	$C_3^4(0.11)^3(0.89)^1 = 0.0047$
4	$C_4^4(0.11)^4(0.89)^0 = 0.0001$

b)

$$E(X) = np = 4(0.11) = 0.44$$

$$\text{Var}(X) = np(1 - p) = 4(0.11)(0.89) = 0.3916$$

$$\text{standard deviation} = \sqrt{0.3916} = 0.6258$$